

xiwang@hms.harvard.edu
(858) 203-8221
Personal Site: samwang.net

Xi (Sam) Wang

EDUCATION

Harvard Medical School

Boston, MA

M.S. in Biomedical Informatics, GPA: 4.0/4.0

Aug 2022 – May 2024

- **Concentrations:** AI/Machine-Learning and Cancer Genomics
- **Coursework:** AI/Machine-Learning in Healthcare, Clinical Informatics, Single-cell Analysis, Statistics, Deep Learning for Biomedical Data Science, Animal Ethics, Launching Technology Ventures, Precision Medicine.

University of California – San Diego

La Jolla, CA

B.S. in Bioengineering: Bioinformatics, GPA: 3.84/4.0 (CS GPA: 3.88/4.0), Cum Laude

Sep 2018 – June 2022

- **Concentrations:** Computer Science and Cancer Genomics
- **Coursework:** Bioinformatics Techniques, Organic Chemistry, Molecular Biology, Vector Calculus, Advance Computer Science Engineering (R, Python, Java, C, C++), Differential Equations, Biology Lab.

PUBLICATIONS

- **Xi Wang**, Simona Cristea, evoCancerGPT: Generating Cancer Progression Using Single-Cell RNA Sequencing Data in a Zero-Shot Manner, October 2024.
Submitted to The 29th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2025). (Manuscript available at: <https://tinyurl.com/2ap37aks>) (Hyperlink to manuscript)
- Marcos Díaz-Gay, Raviteja Vangara, Mark Barnes, **Xi Wang**, S M Ashiqul Islam, Ian Vermes, Stephen Duke, Nithish Bharadhwaj Narasimman, Ting Yang, Zichen Jiang, Sarah Moody, Sergey Senkin, Paul Brennan, Michael R Stratton, Ludmil B Alexandrov, Assigning mutational signatures to individual samples and individual somatic mutations with SigProfilerAssignment, Bioinformatics, Volume 39, Issue 12, December 2023, btad756, <https://doi.org/10.1093/bioinformatics/btad756>. (Hyperlink to manuscript)

CONFERENCES

- **Xi Wang**, Simona Cristea, Generating Cancer Progression and Cell State Dynamics Using Single-Cell RNA Sequencing Data with evoCancerGPT, The 18th Program in Quantitative Genomics (PQG Conference 2024), October 2024 (Poster). *Award: Stellar Abstract Award.* (Poster available at: <https://tinyurl.com/2vvue6ku>). (Hyperlink to poster)
- **Xi Wang**, Marcos Díaz-Gay, Ludmil Alexandrov, Mutational Signature Assignment Benchmark, The Annual 35th Undergraduate Research Conference at UC San Diego, May 2022 (Presentation).

RESEARCH EXPERIENCE

Cristea Lab, Hale Family Center for Pancreatic Cancer Research

Boston, MA

Computational Biologist

July 2024 – Current

- Independent project on developing AI/ML transformer decoder-only large foundation language models for inferring and generating cancer evolution in single-cell transcriptomics data under Dr. Simona Cristea, PhD.
 - * Aggregating 6.5 million cells of publicly available scRNAseq data from CellXGene covering 11 cancer types from primary and metastatic patients.
 - * Constructing and pre-training decoder-only transformer foundation models with custom tokenizers and embedding methods with continuous values for generation of cancer cell evolution progression, with each cell as a token.
 - * Given the continuous expression values of a sets of cells in normal stage, the pre-trained decoder can generate the continuous expression values of "future" cells in primary and metastatic stages in a zero-shot manner.
 - * Independent validations show that the evoCancerGPT is able to generate smooth trajectory in increasing pseudotime along key proliferation marker genes.
 - * Fine-tuning the model with paired temporal scRNAseq data to reconstruct biological insights and evaluate model's generation ability.
 - * Exploring scaling law and benchmarks against based-line LSTM and fine-tuned Llama3.1-8B.

Wolpin-Nowak Lab, Dana-Farber Cancer Institute

Boston, MA

Computational Biologist

July 2024 – Current

- Working with Dr. Brian Wolpin, MD, MPH, Dr. Jonathan Nowak, MD, PhD, and Dr. Andrew Aguirre, MD, PhD.
- Analyzing biomarkers and tumor micro-environments of pancreatic cancer using spatially resolved single-cell RNA sequencing data (CosMx 1000 Panels and 6000 Panels).
 - * Collaborating on one of the largest spatial single-cell RNA sequences study on pancreatic cancer with over 130 samples being sequenced under 5 different methods (CosMx, GeoMx, Visium, Xenium, and COMET) from three leading cancer institutes (Dana-Farber Cancer Institute, MD Anderson Cancer Center, and Memorial Sloan Kettering Cancer Center under Break Through Cancer) .
 - * Leading major computational analyses and constructing standardized pipelines for cell type annotations and tumor subtyping analysis, such as Basal vs. Classical for tumor cells, and other more granular subtypes. Validating results with spatial contextualization of each cells.
- Exploring cell-cell communications between PDAC cancer cells of different cell states (Basal vs. Classical) and their surrounding Cancer Associated Fibroblasts (CAF) with CosMx spatial transcriptomics.
Manuscript in preparation (exp. December 2024)
- Quantifying the effects of Paricalcitol in PDAC patients with GeoMx spatial transcriptomics (clinical trial).
Manuscript in preparation (exp. December 2024)

Liu Lab, Dana-Farber Cancer Institute

Boston, MA

Graduate Student Researcher

Nov 2022 – May 2024

- Developed novel deep neural networks to infer melanoma tumor ploidy, purity, cancer cell fractions and heterogeneity from tumor-only targeted panel sequencing data under the mentorship of Dr. David Liu, MD, MPH, MS.
 - * Data cleaning and sub-sampling of TCGA Melanoma whole exome-sequencing into targeted panel sequences.
 - * Variant Allele Frequency (VAF) + mutational status of targeted genes to predict tumor purity.
 - * Read depth and mutational status of targeted genes to predict tumor ploidy.
 - * Statistical equations for Cancer Cell Fraction (CCF) and heterogeneity inferences.

Alexandrov Lab, University of California, San Diego

La Jolla, CA

Undergraduate Bioinformatics Researcher

May 2021 – Aug 2022

- Independent project Comprehensive Benchmarking of Mutational Signature Assignment Tools under the supervision of Dr. Ludmil B. Alexandrov, PhD.
 - * Benchmarked of 20+ different mutational signature assignment tools based on different programming languages and mathematical algorithms (extensive use of Conda environments and Git version control).
 - * Data analysis and benchmarking of the tools considered using synthetic datasets and various qualitative and quantitative metrics (extensive experience in AWS).
 - * Profiled, upgraded, and tested SigProfilerSingleSample, the assignment tool developed by the Alexandrov lab.

Masuda Lab, University of California, San Diego

La Jolla, CA

Staff Research Associate

May 2022 – Aug 2022

- Project on intervertebral disc restoration using the rabbit model with Dr. Koichi Masuda, MD.
 - * Performed animal pre-operational, surgical, and post-operational procedures on over 100 rabbits).
 - * Executed 2D x-rays analysis and 3D micro-CT post-harvesting analysis.

Cartilage Tissue Engineering Lab, University of California, San Diego

La Jolla, CA

Undergraduate Researcher

Apr 2019 – Aug 2022

- Senior Design project on Surgical Restoration of the Anterior Cruciate Ligament and Knee Biomechanics.
- Quantification of Marine Animal Teeth Enamel Structure with micro-CT.

PROJECTS

- **RxPlorer, Harvard University:** RxPlorer provided specialized pre-trained Large Language Model that performs meta-analysis of current research in the biomedical and clinical field with information that is dynamically updated with sanity and citation checks. Fine-tuned Llama2 (by Meta) with QLoRA, utilized Chat GPT3.5 (OpenAI), implemented quantization of LLM, created web-frontend and server-backend with monitoring system through WhyLab.
- **Yogurt World (product: yogurtworlducsd.com):** Using HTML, designed a professional website for UCSD students' favorite yogurt place on campus. Designed the whole website using Figma and then implement the site with HTML on GitHub.

- **Android App– Zoo Seeker:** Designed an Android Java-based app that allows users to search animal exhibits and show the directions to exhibits with the shortest combined distances, using the Traveling Sales Man algorithm. Integrated auto-correction and voice-recognition for users to search for animals.
- **IOS App– UCSD Dining Hall Review (<https://github.com/xiw020-Sam/UCSDDiningHallReview>):** Designed an IOS Swift-based app that lets users to log-in, post reviews, and comment on reviews for various UCSD dining halls.

SKILLS

AI/ML: Experienced with various ML architectures, custom Neural-Network/Transformer constructions and pre-training with Torch and/or Tensorflow, LLM fine-tuning, and deep neural network fine-tuning.

Programming Languages: Advance knowledge in data structure in R, Java, C, C++, Python, Bash, SnakeMake.

Programming Technologies: Git, Node.js, React, MySQL, MongoDB, Flask, AWS, Docker, Google Cloud Service, Linux, HTML/CSS.

Bioinformatics Techniques: Experienced with bioinformatics pipeline development and analysis using whole-genome, whole-exome, and targeted panel sequences. Experienced in single-cell and spatial transcriptomics analysis.

Languages: Native proficiency in English and Mandarin Chinese, conversational proficiency in Cantonese.

SERVICE/LEADERSHIP

Chinese Union (NPO), University of California, San Diego

La Jolla, CA

Vice President of Operation

Sep 2018 – Mar 2021

- Led a team of 10 experienced Chinese Union Executive-Board members to supervise all CU activities at UCSD.
- Managed all 26 Chinese Union chapters across the U.S.; Assisted the foundation of three additional chapters of Chinese Union at UIUC, SBCC, Chapman University, SJSU, Cal Poly SLO, and OSU.
- Increased the total involvement to more than 100 chapters members.
- Planned and organized student events with thousands of attendees; Monitored and improved programs in all its lifecycles.
- Initiated new programs to help thousands of international students to immerse into UCSD effectively and efficiently.

UCSD Health and Massachusetts General Hospital

La Jolla, CA and Boston, MA

Healthcare Volunteer

Jul 2021 – Jun 2023

- Volunteered for over 130 hours
- Emergency Department, Surgical Oncology In-Patient Unit, and PRE-OP/PACU Unit.
 - * Shadowed emergency physicians' operations and provided patient assistant.